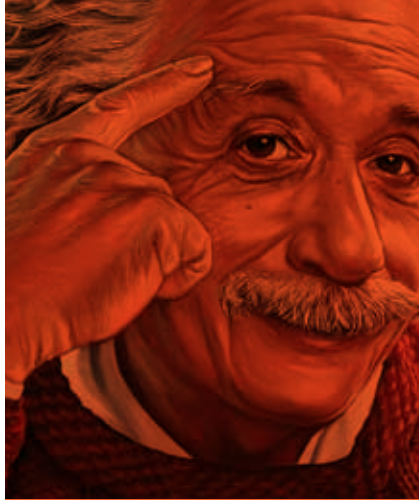




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you stop
learning,
you start
dying.

—Albert Einstein



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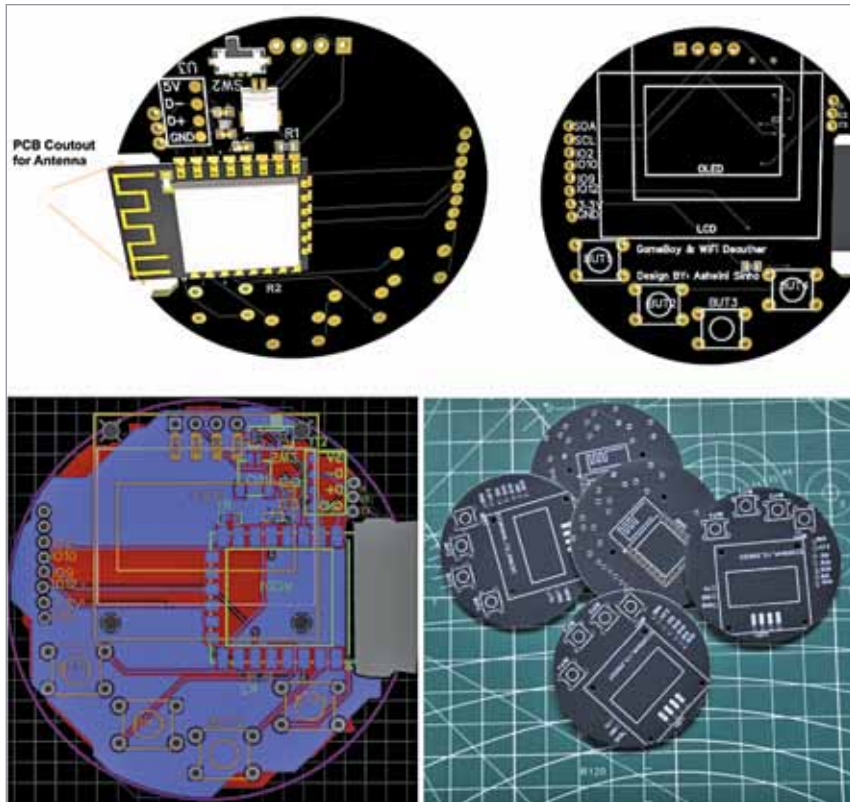


Fig. 7: PCB design showing the antenna cutout on the board and the manufactured PCB

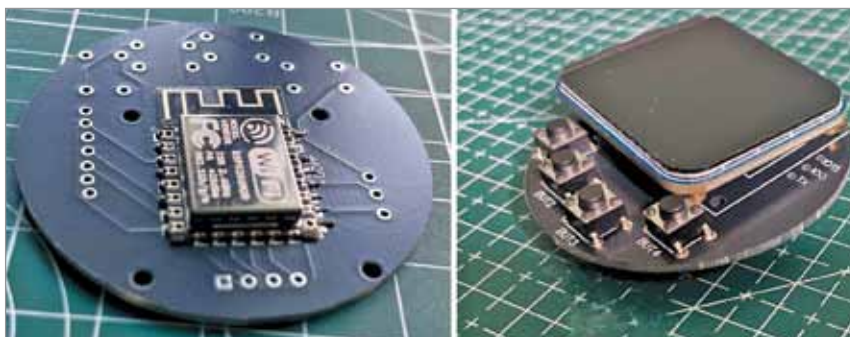


Fig. 8: ESP12F chip SMT mounted on the PCB (left) and complete components assembled on the PCB (right)

footprint and discrete components; the OLED header is on the top side. SPI headers for a colour display are located to the left, and tactile buttons sit below the display. When Wi-Fi is required, the ESP12F antenna must be positioned outward with a PCB cutout to ensure optimal signal performance and reduce interference from surrounding components. A revised layout includes an antenna cutout and proper module placement. Fig. 7 illustrates the PCB layout with the antenna cutout as well as the manufactured PCB.

SMT soldering

After PCB fabrication, perform SMT soldering, install the tactile buttons, and mount the display. The device is then ready for testing. Fig. 8 shows the ESP12F SMT placement and the assembled board.

Software

Custom games can be developed and deployed to the device. Example recreations include Flappy Bird, Pong, and Tetris. Source code and resources are available via the QR